Interfacing Protocols in Embedded Systems

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Introduction to Communication in Microcontrollers

- In Microcontrollers, communication is simply exchanging data between two or more microcontrollers.
- There are many protocols for communication, but all of those are based on either
  - serial communication
  - parallel communication.

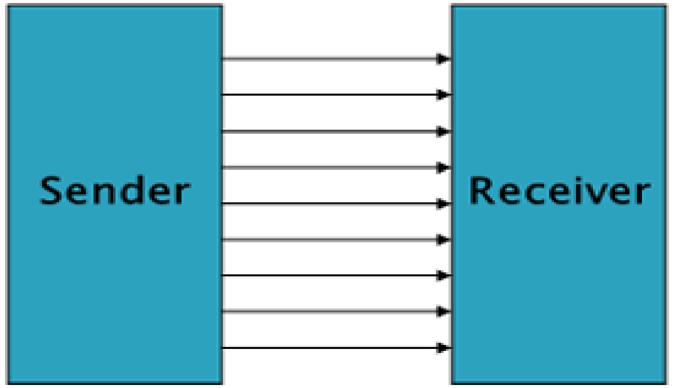
Serial communication

- In serial communication the data bits are transmitted serially one by one i.e. bit by bit on single communication line
- It requires only one communication line rather than n lines to transmit data from sender to receiver.
- Thus all bits of data are transmitted on single lines in serial fashion.
- Less costly.
- Long distance transmission.

Example: Telephone

Parallel communication

- In parallel communication, all the bits of data are transmitted simultaneously on separate communication lines.
- Used for shorter distance.
- In order to transmit n bit, n wires or lines are used.
- More costly.
- Faster then serial transmission.
- Data can be transmitted in less time.



Example: Printers and Hard disk

Serial communication uses two methods:

- Asynchronous
- Synchronous.

Asynchronous:

- ➢ Transfers single byte at a time.
- ➢ No need of clock signal

Example: UART (Universal Asynchronous Receiver Transmitter)

Synchronous:

- ➢ Transfers a block of data at a time.
- ➢ Requires clock signal

Example: SPI (Serial Peripheral Interface)
I2C (Inter Integrated Circuit)

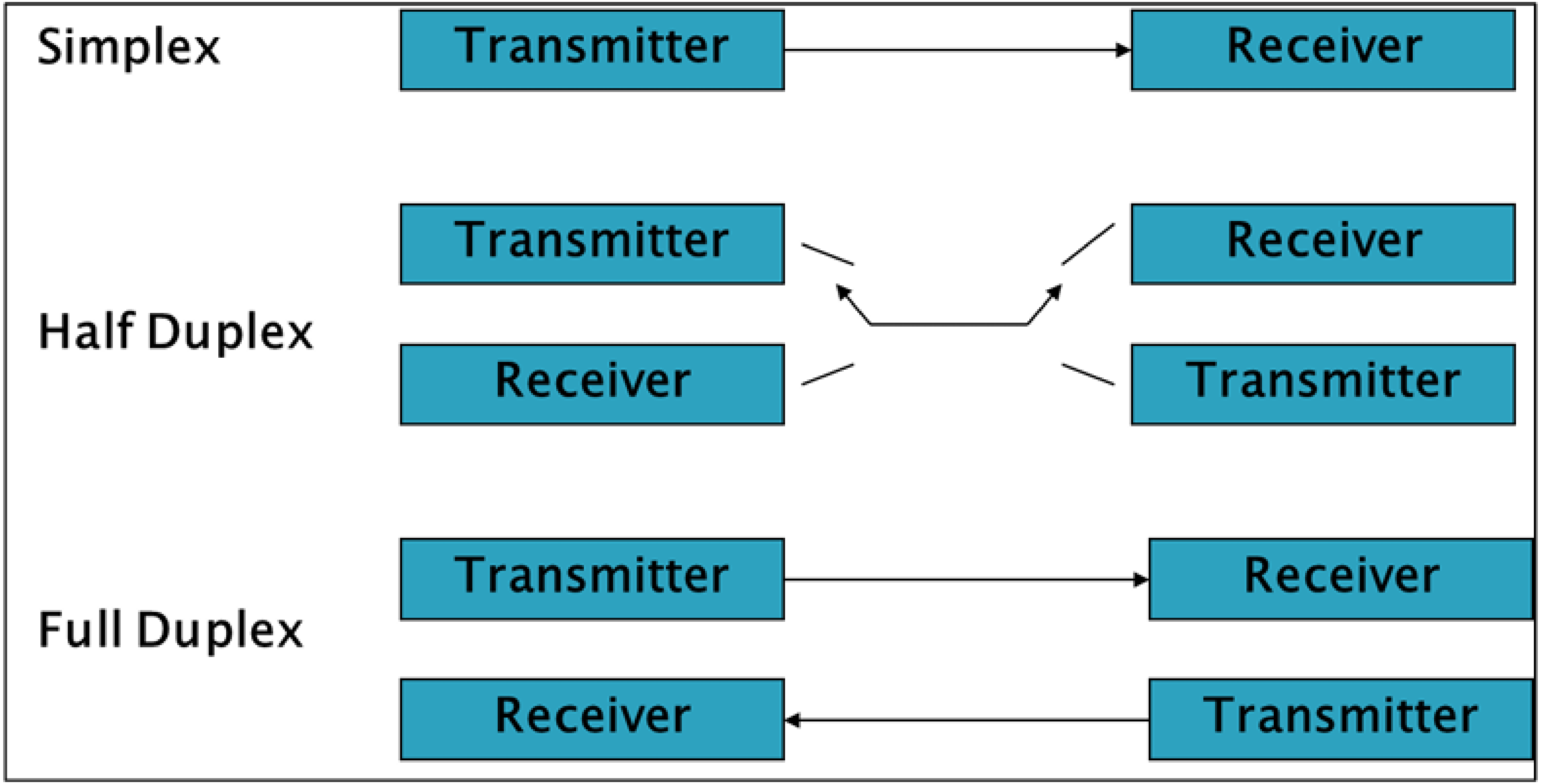
Data Transmission

In data transmission if the data can be transmitted and received, it is a **duplex transmission.**

Simplex: Data is transmitted in only one direction i.e. from TX to RX only one TX and one RX only

Half duplex: Data is transmitted in two directions but only one way at a time i.e. two TX's, two RX's and one line

Full duplex: Data is transmitted both ways at the same time i.e. two TX's, two RX's and two lines

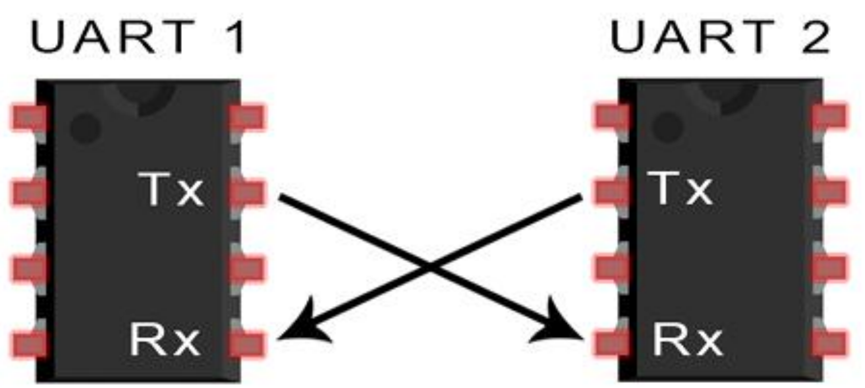

Simplex
Transmitter
Receiver
Half Duplex
Transmitter
Receiver
Receiver
Transmitter
Full Duplex
Transmitter
Receiver
Receiver
Transmitter

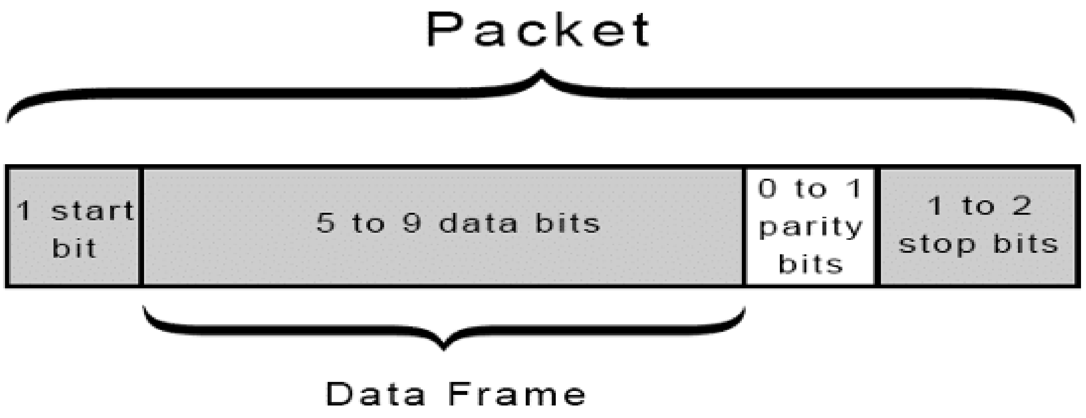
Various Serial Communication Protocols

Serial Protocol	**Synchronous /Asynchronous**	**Type**	**Duplex**	**Data transfer rate (kbps)**
UART	Asynchronous	peer-to-peer	Full-duplex	20
I2C	Synchronous	multi-master	Half-duplex	3400
SPI	Synchronous	multi-master	Full-duplex	>1,000

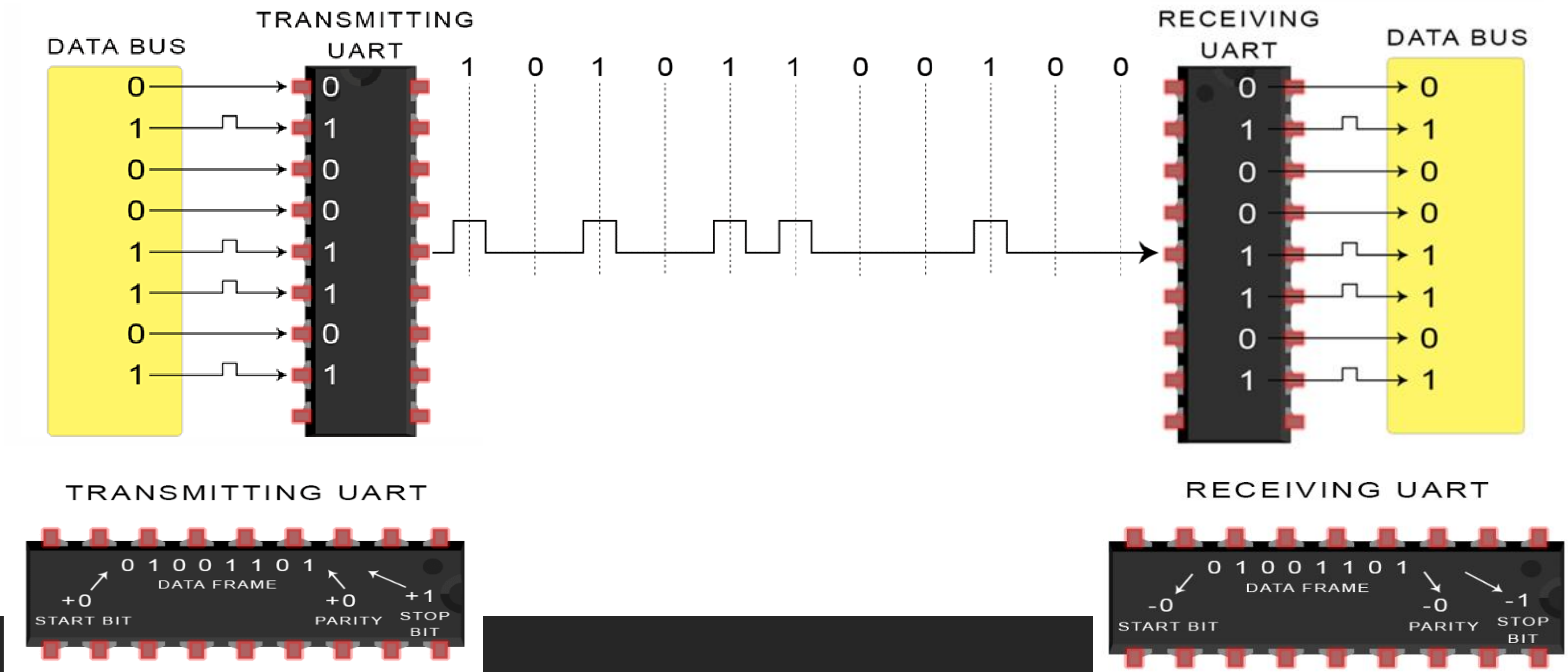
UART
(Universal Asynchronous Receiver Transmitter)

- It is basically **a hardware** that convert **parallel data** into **serial data**.
- UART adds specific bit to the data packet instead of the clock signal.
- UART transmitted data is organized into packets. Each packets contains 1 start bit, 5 to 9 data bits (depending on the UART), an optional parity bit, and 1 or 2 stop bits.



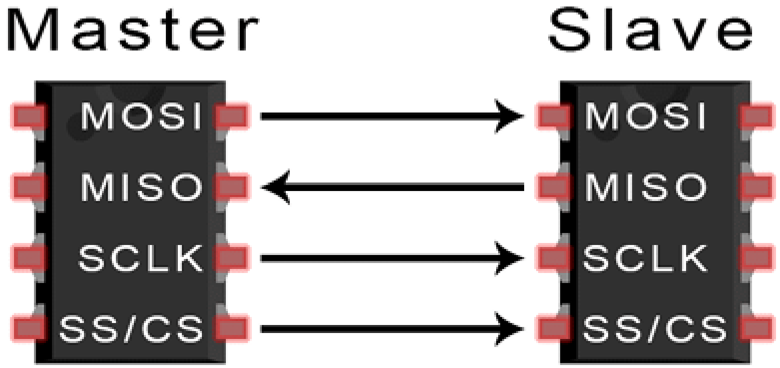


UART Working



SPI
(Serial Peripheral Interface)

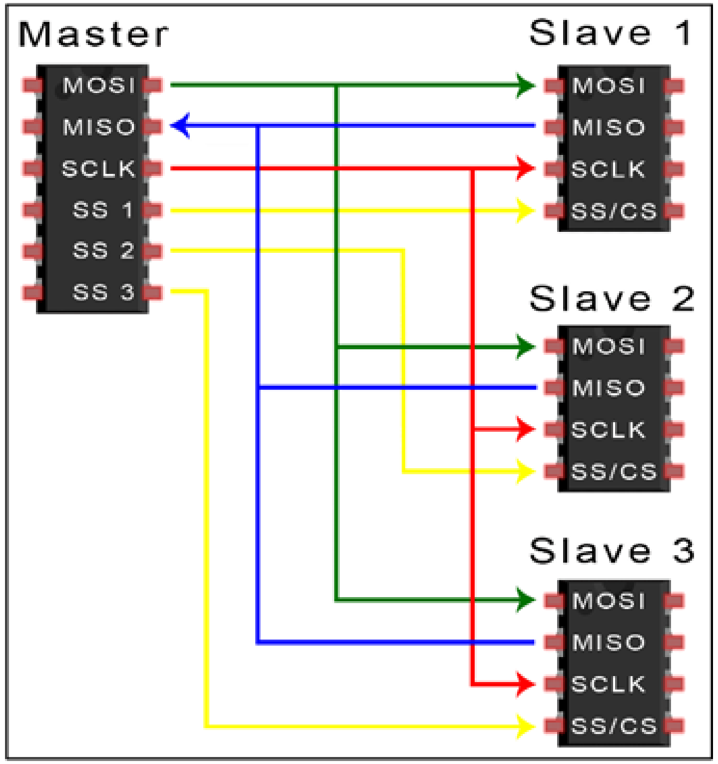
- Common Synchronous communication protocol used by many devices. For example: SD card, RFID card reader, and 2.4GHz wireless transmitter/receivers.
- Sending data without interruptions.
- Master/Slave relationship, where master controls data transfer.



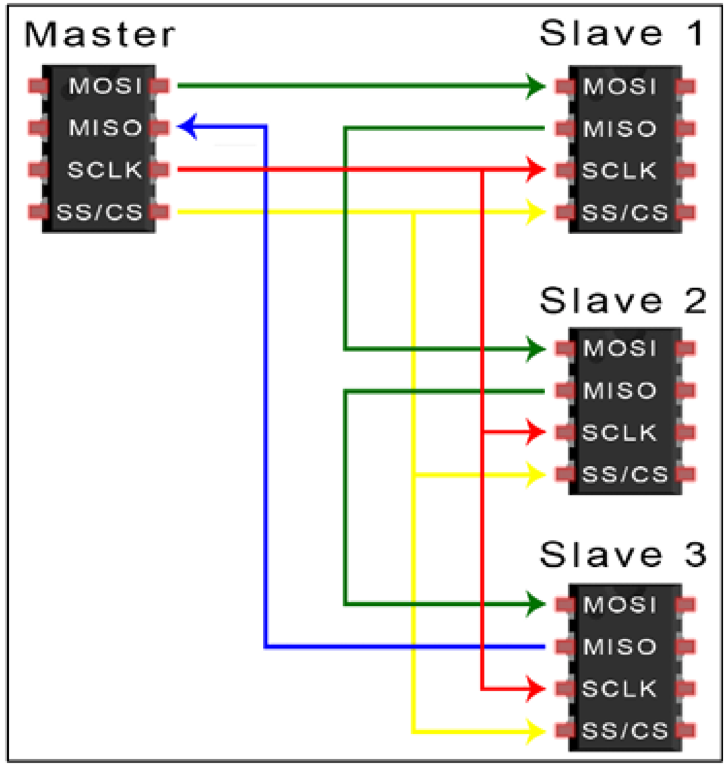
SPI

- Designed to transfer data between various IC chips, at very high speed, hence short bus connections is used to maintain stability.
- Data and control lines of the SPI are:
  - Master Out Slave In (MOSI) also called Serial Data In (SDI).
  - Master In Slave Out (MISO) also Called Serial Data Out (SDO).
  - Serial Clock (SCLK or SCK): generated by the master for Synchronization.
  - Slave Select (SS) on the master and Chip Select (CS) on the slave.

SPI



SPI bus in independent slave configuration



Daisy-Chained SPI Bus

I2C
(Inter Integrated Circuits)

• Bi-directional 2-wire, Multi Master, Low Bandwidth, and short distance serial communication bus protocol.

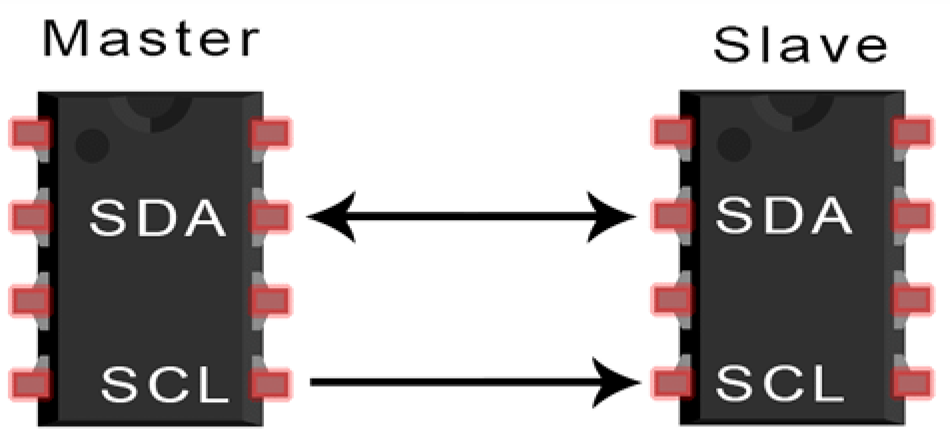
• Has a **7-bit address space** with **16 reserved address**, hence 112 is the maximum number of nodes that can communicate on the same bus.

• The two bi-directional lines are:

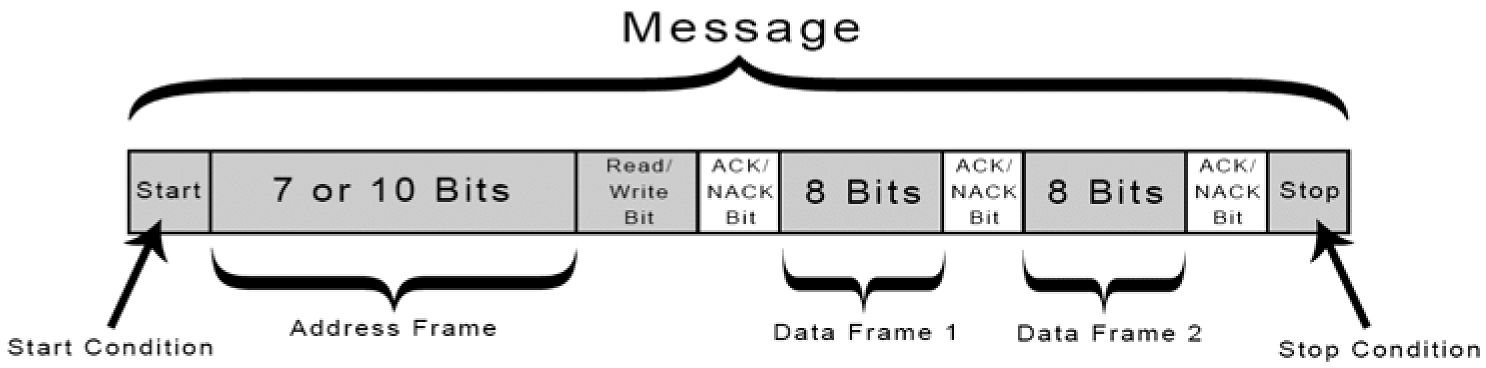
– **SDA**: Serial Data line for data transfer

– **SCL**: Serial Clock line for clock signal and synchronization.

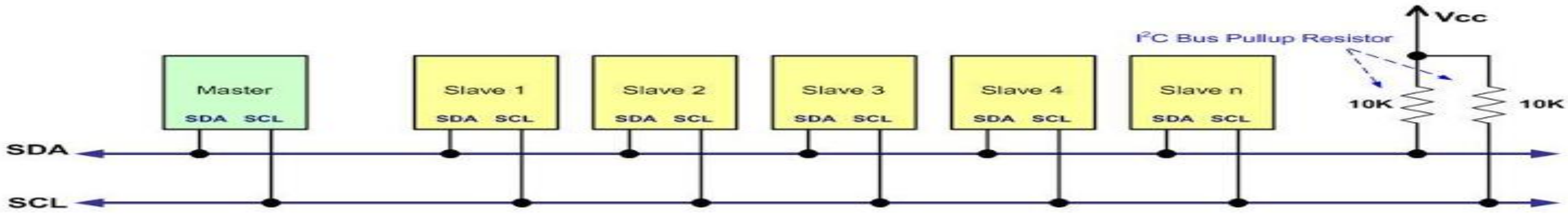
• Typical voltage used are +5V and +3.3V



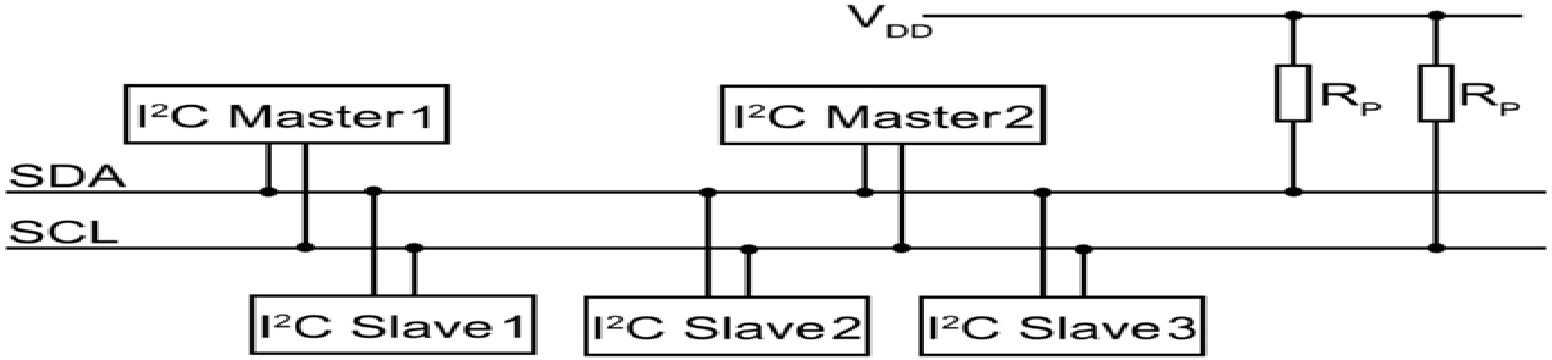
I2C Working



Message
Start
7 or 10 Bits
Read/
Write
Bit
ACK/
NACK
Bit
8 Bits
ACK/
NACK
Bit
8 Bits
ACK/
NACK
Bit
Stop
Start Condition
Address Frame
Data Frame 1
Data Frame 2
Stop Condition



Single Master with Multiple Slaves



Multiple Masters with Multiple Slaves

USB
(Universal Serial Bus)

- It is a set of interfacing specifications for high speed wired communication between electronic systems and peripherals and devices using single standardized interface socket.
- Advantages of USB:

– Low cost.

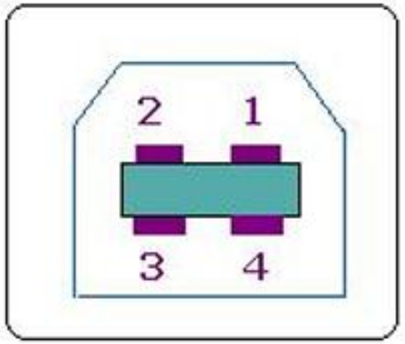
– Expandability.

– Auto configuration.

– Hot swapping.

– Providing power to the bus.

Type A socket



Type B socket

USB Generation Comparison Table

Feature	USB 1.0	USB 1.1	USB 2.0	USB 3.0 (3.1 Gen 1)	USB 3.1 (3.1 Gen 2)
Common Name	Low Speed	Full Speed	Hi-Speed	SuperSpeed	SuperSpeed+
Max Speed	1.5 Mbps	12 Mbps	480 Mbps	5 Gbps (500 MB/s)	10 Gbps (1.2 GB/s)
Release Year	1996	1998	2000	2008	2013
Data Direction	One-way (Half-duplex)	One-way (Half-duplex)	One-way (Half-duplex)	Two-way (Full-duplex)	Two-way (Full-duplex)
Connector Color	White/Black	White/Black	Black	Blue	Teal/Red/Blue
Max Power	2.5W (5V/0.5A)	2.5W (5V/0.5A)	2.5W (5V/0.5A)	4.5W (5V/0.9A)	100W (20V/5A)*
Wire Count	4	4	4	9	9

THANK YOU